Summary of Technical Report: Definition and Diagnosis of Diabetes Mellitus and Intermediate Hyperglycaemia (WHO / IDF)

Executive Summary & Key Recommendations:

Recommendation 1: The current WHO diagnostic criteria for diabetes should be maintained:
? Fasting plasma glucose (FPG) >= 7.0 mmol/L (126 mg/dL) OR
? 2-hour post-load plasma glucose >= 11.1 mmol/L (200 mg/dL).

Recommendation 2: Use the term 'normoglycaemia' for glucose levels associated with low risk of developing diabetes or CVD (< 6.1 mmol/L FPG).

Recommendation 3: Maintain current WHO definition for Impaired Glucose Tolerance (IGT):
? FPG < 7.0 mmol/L (126 mg/dL) AND 2-h post-load glucose >= 7.8 mmol/L (140 mg/dL) and < 11.1 mmol/L (200 mg/dL).

Recommendation 4: Maintain FPG cut-point for Impaired Fasting Glucose (IFG) at 6.1 to 6.9 mmol/L (110?125 mg/dL). (Do not lower to 5.6 mmol/L as proposed by ADA 2003, to avoid 2-3 fold artificial increase in prevalence).

Recommendation 5:
1. Venous plasma glucose is the standard method for measuring and reporting glucose concentrations.
2. Samples should be processed immediately or collected with glycolytic inhibitors on ice-water.

Recommendation 6: The Oral Glucose Tolerance Test (OGTT) should be retained as a diagnostic test because FPG alone fails to diagnose ~30% of cases of undiagnosed diabetes and cannot detect IGT.

Recommendation 7: HbA1c is not recommended as a sole diagnostic test in resource-limited settings due to lack of global standardization and confounding by anaemias/haemoglobinopathies.

Appendix: Summary of Diagnostic Criteria Across WHO Revisions

Normal: FPG < 6.1 mmol/L (110 mg/dL), 2-h post-load < 7.8 mmol/L (140 mg/dL).
Diabetes Mellitus: FPG >= 7.0 mmol/L (126 mg/dL) OR 2-h post-load >= 11.1 mmol/L (200 mg/dL).
Impaired Glucose Tolerance (IGT): FPG < 7.0 mmol/L AND 2-h post-load 7.8 to 11.0 mmol/L.
Impaired Fasting Glucose (IFG): FPG 6.1 to 6.9 mmol/L AND 2-h post-load < 7.8 mmol/L.